

# Kyle Tran

MECHANICAL ENGINEERING • ROBOTICS MINOR • NORTHEASTERN '29

kylegtran2007@gmail.com  
ka1e27.github.io/Portfolio  
linkedin.com/in/kyle-tran-06a18b277  
github.com/ka1e27  
Boston • Burlingame, CA

Seeking a Summer 2027 co-op in hardware and robotics. I build robots, autonomous vehicles, and embedded systems — drawn in CAD, machined and printed, wired and coded, then tested until they hold up: two years of competitive combat robotics, and an autonomous survey boat built from scratch.

|                                     |                         |                                |                                |
|-------------------------------------|-------------------------|--------------------------------|--------------------------------|
| 4th <small>NATIONALLY, 2024</small> | 3.87 <small>GPA</small> | 10 <small>ROBOTS BUILT</small> | 16 <small>TEAM PODIUMS</small> |
|-------------------------------------|-------------------------|--------------------------------|--------------------------------|



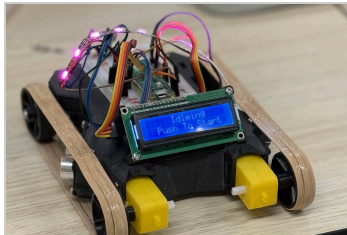
D.R.E.A.M.S v1 on the water



Creators' Showcase, NU London



Combat robots — Team DHM



Maze-navigating robot

## PROJECTS

### Autonomous Environmental Survey Boat — D.R.E.A.M.S

Jan 2026 — Present

A twin-hull autonomous survey boat, co-founded with Matthew Hong. V1, built in one semester in London, navigated to GPS waypoints on its own at St Katharine Docks and winched a sensor pod to the bottom to sample. V2 added a Raspberry Pi 5, stereo vision and our first custom PCB, but bench tests exposed flaws in its motor control, power and safety. V3, in build, pairs a Pi 5 with a Teensy 4.1 on a custom 4-layer carrier board. I lead the hardware; the software was built with AI coding agents.

SolidWorks • ASA + fiberglass hull • KiCad 4-layer PCB • Pi 5 + Teensy 4.1 • CAN • VESC • RS-485 • YOLOv8 • GPS/IMU • 4G

### Combat Robotics — Team DHM

2023 — 2025

Designed and built around ten robots that competed across five weight classes — weapon and drivetrain design, titanium and steel armor, composite layup, and iteration driven by what broke in the arena. 4th nationally in Plastic Antweight (2024), U.S. rankings of top-15, top-30, and top-35, and 4 firsts and 8 seconds at regional events; the club took 16 podiums over 5 events. The newest Achilles walks on a single-cam shuffler drive I designed.

Onshape • CNC-machined aluminum (outsourced) • laser-cut Ti / steel • brushless drive • FDM • DFM to a weight limit

### Adaptive Compliant Robotic Arm

2026 — in design

A five-axis arm with a  $\pm 0.25^\circ$  static target and a compliant TPU gripper that reads its own grip force. The base, shoulder and elbow run cycloidal gearboxes (14:1 to 22:1) with output-side magnetic encoders, and all three pass every check in a gearbox simulator I specified and built with AI coding agents. Nothing is built yet: a printed prototype comes first, then 7075 discs laser-cut by SendCutSend and finished on a Bantam PCB mill that I run.

Cycloidal drives • gearbox simulator • closed-loop steppers • output-side encoders • TPU gripper • tolerance budgeting

### Autonomous Maze-Navigating Robot

Oct — Dec 2025

A 3D-printed tank-drive robot, built with Matthew Hong for the Cornerstone of Engineering course at Northeastern, that completed all three challenge courses. I did the CAD, the printing and the MicroPython firmware: proportional steering off two angled ultrasonic sensors, with asyncio so the LCD and LEDs never stall the motors.

Onshape • 3D printing • MicroPython • asyncio • RP2040 • ultrasonic + IR • I2C LCD

### Miscellaneous Builds

2023 — Present

10+ custom ceramics tools designed in CAD, printed, and open-sourced for anyone to print; 3D-printed RC aircraft with carbon-fiber spars; a full sewing machine reverse-engineered in CAD; and printed climbing hardware.

CAD to FDM • published on MakerWorld • github.com/ka1e27

# Experience & Skills

## EDUCATION

**B.S. Mechanical Engineering** · Minor in Robotics Sep 2025 – May 2029 (exp.) · GPA 3.87  
Northeastern University · Boston, MA  
Cornerstone of Engineering I · Statics · Thermodynamics · Materials Science · Probability & Statistics

## LEADERSHIP EXPERIENCE

**President** Oct 2023 – Jun 2025  
Combat Robots Club, Burlingame

Ran workshops on CAD, soldering, physics, materials, and 3D printing; handled sponsorships and club finances. The team placed on the podium 16 times – 4 firsts, 7 seconds, 5 thirds – across 5 events.

**Ceramic Tools R&D** Oct 2023 – May 2025  
Burlingame High School · Ceramics Program  
Modeled, printed, and published 10+ ceramics tools – trimming tools, ribs, wire guides, stamps – free on MakerWorld.

**Secretary · Co-founder** Jan 2024 – May 2025  
Burlingame Aeronautics Club  
Helped found the club and taught the physics of flight – aerodynamics, lift, drag – then put it into practice designing 3D-printed RC aircraft.

**Subsystem Lead** Sep 2021 – May 2024  
Iron Panthers FRC #5026  
Designed the spindexter mechanism for the competition robot, working with 10+ team members and 5 professional mentors through design reviews and manufacturing.

## SKILLS

DESIGN & CAD  
SolidWorks · Onshape · Fusion 360 · mechanism design · DFM

ELECTRONICS & FIRMWARE  
KiCad · custom PCB · Raspberry Pi · RP2040 · MicroPython · soldering

SOFTWARE & AI  
Python/asyncio · MATLAB · Java · AI coding agents · simulation-first development

FABRICATION  
2.5D CNC machining · CNC router · PCB mill · waterjet · laser cutter · FDM printing · composite layup (FG/CF)

ROBOTICS & AUTONOMY  
Stereo camera rigs · GPS + IMU fusion · VESC drive · 4G telemetry · state machines

LEADERSHIP  
Club president · workshops · sponsorships · mentoring

## AWARDS

|     |                                           |                               |
|-----|-------------------------------------------|-------------------------------|
| 4th | Robot Combat League National Championship | 1 lb plastic antweight · 2024 |
| 1st | CROS – March Madness                      | 1 lb antweight · 2024         |
| 1st | BCRC Bell Brawls IV                       | 3 lb beetleweight · 2024      |
| 1st | Spartan Bell Slugfest                     | 1 lb plastic antweight · 2024 |
| 1st | NorCal Robotics Expo                      | 1 lb plastic antweight · 2023 |
| 2nd | Eight further second-place finishes       | ant / beetleweight · 2024     |